南京信息工程大学

Real-Time Object Detection Using YOLO and Its Variants: An Implementation Study

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Abstract

The precise identification and localization of sugarcane stem nodes is critical for intelligent agricultural practices such as automated seed cutting and precision planting. However, existing detection methods face substantial challenges in computational efficiency, real-time performance, and deployment feasibility in resource-constrained agricultural environments.

This study presents a lightweight detection methodology based on the YOLOv8_n architecture for real-time sugarcane stem node detection. A field-collected dataset of 450 images across two commercially significant varieties was constructed, and systematic hyperparameter optimization with advanced data augmentation was implemented to enhance model generalization.

Experimental results demonstrate exceptional performance: precision of 0.973, recall of 0.958, mean Average Precision (mAP) of 0.974, single-frame inference time of 19.80 ms, and model size of merely 6.30 MB. Compared to Edgeyolo_S_Coco and Edgeyolo_Tiny baselines, the proposed method achieves higher accuracy while significantly reducing inference time and model size. The method exhibits strong robustness under challenging field conditions and provides a viable technical foundation for deployment on edge devices in intelligent sugarcane production systems.

Keywords

YOLOv8; Real Time Object Detection; Stem Node Detection; Sugarcane; Intelligent Agriculture;

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INTRODUCTION

1.1. Background: The Role of Sugarcane in Global Agricultural Economics

Sugarcane (*Saccharum officinarum*) stands as one of the most critical cash crops in global agricultural economics, serving as the primary raw material for over 70% of the world's sugar production and playing an increasingly vital role in the bioenergy sector. As a cornerstone of agricultural economic development, the promotion of high-quality development within the sugarcane industry is of paramount significance. It directly impacts global sugar supply chains, accelerates the transformation and upgrading of traditional agricultural practices, and fosters high-quality, sustainable economic development in major producing regions [1, 2].

In countries like China, the sugarcane industry is undergoing profound structural changes. Historically reliant on extensive, labor-intensive farming models, the sector now faces mounting pressures from rising labor costs, an aging agricultural workforce, and the urgent need for modernization. To maintain competitiveness and ensure food and energy security, agricultural technological innovation has emerged as the key driving force for the modernization of the sugarcane industry. The transition from conventional farming to precision agriculture is no longer merely an option but an agronomic and economic imperative. Central to this transformation is the automation of critical field operations, particularly those related to seed preparation, precise planting placement, and optimized harvesting, all of which hinge on the accurate identification of specific morphological features of the crop, most notably the sugarcane stem nodes.

1.2. Problem Statement: Challenges in Manual vs. Automated Stem Node Detection

The precise identification and localization of sugarcane stem nodes are foundational to intelligent agricultural operations. Stem nodes dictate the optimal cutting points for seed preparation and determine the planting depth and orientation, which directly influence germination rates, yield improvement, and overall economic profitability.

Currently, field operations in sugarcane cultivation still rely heavily on manual labor. Manual seed preparation and planting are not only characterized by low work efficiency and high physical demand but also suffer from inherent human inconsistencies. These manual processes cannot guarantee the uniformity and accuracy required for large-scale, high-yield agricultural production [3]. Consequently, the inability to standardize these operations creates a significant bottleneck in scaling sugarcane production efficiently.

While there is an urgent need for automation, the accuracy and efficiency of current automated sugarcane stem node detection methods remain inadequate for widespread commercial application. Designing an efficient, robust, and highly accurate sugarcane stem node detection system represents a critical bottleneck facing the intelligent management of sugarcane orchards. The core challenge lies in developing a system that can operate reliably in unstructured, dynamic outdoor environments characterized by variable lighting, occlusion, overlapping stems,

and complex background clutter, all while maintaining the speed and efficiency required for real-time agricultural machinery operations.

1.3. The Shift Toward Smart Agriculture and Computer Vision

To address the limitations of manual labor, **computer vision and image processing technologies** have gradually been integrated into sugarcane planting and harvesting research. This shift toward smart agriculture has proven effective in alleviating societal labor shortages and significantly improving operational efficiency [4].

Early research in this domain primarily relied on traditional machine learning and image processing techniques. For instance, preliminary research by Moshashai et al. [5] utilized grayscale image threshold segmentation for stem node recognition. Similarly, Dhingra Gittaly et al. [6] explored methods based on local mean and the H (Hue) component in the HSV color space, achieving a recognition rate of 90.77% with an average processing time of 0.48 seconds. Chen Wen et al. [7] applied Support Vector Machines (SVM) to process S and H component images in the HSV color space, achieving average recognition rates of 94.11% for stem node number and 91.52% for position after clustering analysis. Furthermore, Wang Weiwei et al. [8] investigated hyperspectral imaging technology, extracting specific feature bands to recognize stem nodes with positioning errors of less than 0.9 mm and 2.4 mm.

While these traditional methods achieved notable milestones, they suffer from inherent limitations. They rely heavily on manually engineered features, which results in poor robustness and weak generalization ability when deployed in the complex, highly variable environments of actual sugarcane fields. Consequently, their recognition performance degrades significantly under real-world conditions.

In recent years, deep learning technology has revolutionized agricultural computer vision, demonstrating superior capabilities in automatic feature extraction and object detection, especially in complex environments [9–14]. Researchers have increasingly turned to Convolutional Neural Networks (CNNs) and the YOLO (You Only Look Once) family of models. For example, Xu Hongzhen et al. [15] improved YOLOv3 to achieve **90.38% accuracy in 28.7 ms**. Sangaiah Arun Kumar et al. [16] enhanced YOLOv4, reducing single-frame recognition time to approximately 6 ms with **98.68% accuracy**. Other studies have explored improvements to LeNet-5 [17], YOLOv5 [18, 19], and the integration of attention mechanisms like CBAM [20].

However, a critical research gap remains. The vast majority of existing deep learning studies focus predominantly on maximizing algorithmic accuracy, often at the expense of real-time operation and model lightweighting. These high-complexity models require substantial memory and computing resources, typically necessitating high-performance desktop GPUs for inference. This requirement directly contradicts the current trend toward the miniaturization of agricultural edge devices and significantly increases the cost burden of mechanization in rural planting environments. Therefore, the core challenge in modern sugarcane production equipment is how to drastically reduce the computational cost and volume of the detection model while maintaining, or even enhancing, system performance and accuracy.

1.4. Research Objectives and Core Contributions of this Study

To bridge the gap between high-accuracy deep learning models and the practical, resource-constrained needs of sugarcane farmers and production equipment, this study proposes an advanced, efficient, and lightweight detection method for sugarcane stem nodes based on the YOLOv8 architecture.
The primary objectives and core contributions of this research are threefold:

1. Comprehensive Dataset Construction in Natural Environments: We designed and executed rigorous field-based image acquisition experiments in structured outdoor scenes. We manually annotated and randomly partitioned the data to construct a high-quality, representative sugarcane image dataset encompassing two commercially significant varieties (New Tai Sugar 22 and Gui Sugar 49), ensuring the model is trained on diverse, real-world agronomic data.
2. Optimization of the YOLOv8 Detection Framework: We utilized the YOLOv8 network as the foundational detection model and systematically optimized its hyperparameter combinations. By leveraging advanced data augmentation techniques (**Mosaic, Mixup, HSV transformation, EMA**) and adaptive anchor box determination via K-means clustering, we achieved an optimal balance between detection precision and computational efficiency.
3. Rigorous Comparative Evaluation of Performance and Complexity: We designed and conducted comprehensive comparative experiments against established baseline models (Edgeyolo_S_Coco and Edgeyolo_Tiny). This evaluation comprehensively assesses not only detection metrics (Precision, Recall, mAP) but also critical deployment metrics, including single-frame inference time and overall model size.

The research results provide robust technical support for the intelligent harvesting and planting of sugarcane. By delivering a model that is both highly accurate and exceptionally lightweight, this study directly addresses the practical needs of modern sugarcane production equipment, paving the way for reduced operational costs, improved efficiency, and the broader realization of intelligent agricultural production systems.

The remainder of this paper is organized as follows. Section 2 reviews related work on traditional and deep learning-based detection methods. Section 3 details the proposed methodology, including dataset construction and the YOLOv8 architecture. Section 4 presents the experiments, results, and discussion. Finally, Section 5 concludes the paper and outlines future work.

Related Work

The automated detection and localization of sugarcane stem nodes represent a complex computer vision challenge that has evolved significantly over the past two decades. This section provides a comprehensive review of the methodological progression in this domain, transitioning from traditional, manually engineered image processing techniques to modern, data-driven deep learning architectures. By critically analyzing the strengths and limitations of existing literature, this review establishes the theoretical foundation and identifies the critical research gaps that this study aims to address.

2.1. Traditional Machine Learning and Image Processing Techniques

The earliest attempts at automating sugarcane stem node detection relied heavily on classical computer vision and traditional machine learning paradigms. These methods generally followed a sequential pipeline: image acquisition, pre-processing, manual feature extraction, and classification or segmentation.

Thresholding and Grayscale Segmentation:

Pioneering research by Moshashai et al. [5] explored the feasibility of using machine vision for sugarcane node identification through grayscale image threshold segmentation. Their methodology was predicated on the assumption that stem nodes and internodes exhibit distinct intensity profiles due to morphological ridges and shadow casting. While this approach established a baseline for automated detection, it proved highly susceptible to environmental noise. Variations in solar illumination, canopy shadows, and soil background clutter frequently caused the global thresholding algorithms to fail, resulting in severe over-segmentation or under-segmentation.

Color Space Transformations and Local Mean Methods:

To mitigate the vulnerabilities of grayscale thresholding, subsequent researchers turned to color space transformations. Dhingra Gittaly et al. [6] proposed a segmentation method utilizing the local mean and the Hue (H) component within the HSV (Hue, Saturation, Value) color space. By isolating the chromatic properties of the sugarcane bark and utilizing maximum grayscale values to pinpoint node locations, their system achieved a recognition rate of 90.77% with an average processing time of 0.48 seconds per frame. The transition to HSV was a significant theoretical advancement, as it decouples color information from illumination intensity, offering improved robustness against the harsh lighting conditions typical of outdoor agricultural environments.

Support Vector Machines (SVM) and Feature Engineering:

Further refining the feature extraction process, Chen Wen et al. [7] developed a comprehensive machine vision framework that applied differential processing to both the Saturation (S) and Hue (H) components in the HSV color space. Following feature extraction, they employed a Support Vector Machine (SVM) classifier to delineate stem nodes from internodes. Post-clustering analysis yielded average recognition rates of 94.118% for node enumeration and 91.522% for spatial positioning. The use of SVMs introduced non-linear decision boundaries, allowing for better separation of complex textural features compared to simple thresholding.

Hyperspectral Imaging and Spectral Analysis:

Pushing beyond the visible light spectrum, Wang Weiwei et al. [8] investigated hyperspectral imaging technology for precise node localization. By capturing spectral data via a spectrometer and extracting specific feature bands unique to the biochemical composition of the node tissue, they established a highly accurate recognition model. Their system reported remarkable positioning errors of less than **0.9 mm and 2.4 mm** for the left and right ends of the stem nodes, respectively.

Limitations of the Traditional Paradigm:

Despite these incremental advancements, traditional methods share a fundamental architectural flaw: they rely entirely on manual feature engineering. Designing these features requires extensive domain expertise and is inherently brittle. When confronted with the extreme morphological diversity of different sugarcane varieties, occlusion from overlapping leaves, or adverse weather conditions (e.g., rain, dust), the handcrafted features degrade rapidly, leading to poor generalization and weak robustness in unstructured, real-world field environments.

2.2. Evolution of Deep Learning in Agricultural Object Detection

The advent of Convolutional Neural Networks (CNNs) catalyzed a paradigm shift in agricultural computer vision. Unlike traditional methods, deep learning models perform end-to-end learning, automatically extracting hierarchical feature representations—from low-level edges and textures to high-level semantic concepts—directly from **raw pixel data**.

The application of deep learning in agriculture has expanded rapidly, demonstrating unprecedented success in various crop detection tasks. For instance, Zheng et al. [10] developed an efficient online citrus counting system for large-scale unstructured orchards using UAV imagery and deep learning. Similarly, Li et al. [11] proposed a multi-scale cucumber disease detection method capable of identifying pathogens in complex natural scenes. In paddy agronomy, Ganesan and Chinnappan [12] successfully hybridized ResNet architectures with YOLO classifiers for automated leaf disease recognition, while Li et al. [13] utilized image fusion techniques to detect abnormal behaviors in aquaculture. Furthermore, Boogaard et al. [14] demonstrated the efficacy of multi-view imaging combined with deep learning for robust node detection and tracking in various fruit-vegetable crops.

These cross-domain successes underscore the superior feature extraction capabilities and generalization potential of deep learning. However, agricultural object detection presents unique challenges, including extreme scale variation, dense occlusion, and the strict real-time processing constraints imposed by mobile machinery. Addressing these challenges requires specialized, highly optimized network architectures.

2.3. Review of YOLO Architectures in Crop and Sugarcane Detection

Among deep learning frameworks, the YOLO (You Only Look Once) series has emerged as the dominant paradigm for real-time agricultural object detection. By framing detection as a single-stage regression problem, YOLO models offer an optimal balance between inference speed and spatial accuracy, making them highly suitable for integration into agricultural machinery.

YOLOv3 and Early Adaptations:
Xu Hongzhen et al. [15] were among the first to adapt the YOLOv3 architecture for sugarcane bud detection. By strategically reducing the number of residual structures composed of intermediate convolutional layers, they developed the YOLOv3-CSE network. This structural pruning decreased computational overhead, achieving an accuracy of 90.38% with an average processing time of 28.7 ms per frame. While successful, the model still required significant computational resources, limiting its deployment on low-cost edge devices.
YOLOv4 and Feature Aggregation:
Sangaiah Arun Kumar et al. [16] advanced the state-of-the-art by leveraging an improved YOLOv4 architecture for UAV-based paddy agronomy and crop sensing. Their innovation involved directly passing effective feature layers into an enhanced feature extraction network for path aggregation construction. This improved multi-scale feature fusion, resulting in a single-frame recognition time of approximately 6 ms and an accuracy of 98.68%. This highlighted the critical importance of the "Neck" network in aggregating semantic and spatial information for high-speed detection.
YOLOv5, Lightweighting, and Attention Mechanisms:
The release of YOLOv5 sparked a wave of research focused on model lightweighting and attention mechanisms, directly addressing the hardware constraints of rural farming environments.

- Lightweighting: Xie Zhongjian et al. [18] implemented a sugarcane stem node recognition algorithm based on an improved YOLOv5 model, specifically targeting model compression. They achieved a lightweight deployment method at the cost of only marginal accuracy loss, proving that parameter reduction is viable for agricultural edge computing.
- Oriented Bounding Boxes: Lu Zhiheng et al. [19] recognized that sugarcane stems often appear at oblique angles relative to the camera. They proposed a recognition method based on an improved YOLOv5obb (oriented bounding box) model, achieving 97.1% accuracy by allowing the network to predict rotated bounding boxes that tightly fit the diagonal geometry of the stems.
- Multi-Scale and Anchor Optimization: Qu Zhong et al. [20] improved the YOLOv5 network by integrating a multi-scale prediction structure and utilizing the K-means algorithm to optimize bounding box anchor dimensions, achieving an mAP of 93.8% for large object detection.
- Attention Mechanisms: Dai Jiuxiang et al. [4] optimized YOLOv5 by embedding the CBAM (Convolutional Block Attention Module) and introducing the VarifocalNet loss function. The attention mechanism allowed the network to dynamically suppress irrelevant background noise (like soil and weeds), achieving a recognition accuracy of 89.89%.

Alternative Architectures:
Outside the YOLO family, Bereket Getachew et al. [17] explored the improvement of the classic LeNet-5 network for plant species and bud classification. While achieving a 92% recognition accuracy, the model required 1.2 seconds per image, underscoring the severe

latency penalties associated with using non-optimized, legacy CNN architectures for real-time tasks.

2.4. Identified Research Gaps: The Need for Lightweight, Real-Time, and Robust Models

Despite the significant milestones achieved in the literature, a critical synthesis of existing research reveals a persistent disconnect between algorithmic performance in laboratory settings and the practical realities of agricultural mechanization.

1. The Accuracy-Efficiency Trade-off:

The vast majority of current studies (e.g., [15], [16], [20]) focus predominantly on maximizing mAP and accuracy, often employing complex, parameter-heavy architectures. These high-complexity models demand high-performance desktop GPUs and substantial memory bandwidth. This reliance on expensive hardware directly contradicts the economic realities of sugarcane cultivation, where mechanization must be low-cost and accessible to rural farmers.

2. Neglect of Edge-Deployment Constraints:

There is a distinct lack of research that systematically optimizes for the strict constraints of edge devices. Agricultural edge computers (e.g., embedded systems on tractors) operate under severe limitations regarding power consumption, thermal dissipation, and physical memory. Models exceeding 20-40 MB in size are often impractical for Over-The-Air (OTA) updates and rapid deployment on these resource-constrained platforms.

3. Inadequate Robustness in Unstructured Scenes:

While attention mechanisms (like CBAM [4]) have been introduced, many models still struggle with the chaotic nature of unstructured outdoor scenes. The vibration of seed-cutting machinery, wind-induced motion blur, and severe occlusion require a level of spatial and temporal robustness that many current YOLOv5 variants fail to maintain without sacrificing inference speed.

Conclusion of the Review:

The literature clearly indicates that the core challenge in modern sugarcane production is no longer just achieving high accuracy, but doing so within a strictly lightweight, real-time framework. To bridge this gap, there is an urgent need for a next-generation architecture that inherently balances feature extraction efficiency with aggressive parameter reduction. This study addresses these identified gaps by proposing an advanced, highly optimized detection method based on the YOLOv8 architecture, specifically engineered to deliver state-of-the-art accuracy while maintaining a microscopic model footprint and ultra-fast inference times suitable for autonomous agricultural machinery.

Methodology

3.1. Field Experiment Setup and Image Acquisition Protocol

The foundation of any robust deep learning model lies in the quality, diversity, and representativeness of its training data. To ensure the proposed YOLOv8-based detection method could generalize effectively to real-world agricultural environments, a rigorous, field-based image acquisition experiment was designed and executed.

3.1.1. Sugarcane Varieties and Field Conditions

The data collection was conducted in November 2023 at the Agricultural Machinery Research Institute of the Chinese Academy of Tropical Agricultural Sciences, located in Zhanjiang, Guangdong Province, China (coordinates: 21°10'N, 110°16'E). This location was strategically chosen due to its status as a major sugarcane-producing region, offering authentic, unstructured outdoor field conditions representative of actual farming environments.

To ensure the model could generalize across different morphological characteristics, the dataset was constructed using two commercially significant and visually distinct sugarcane varieties:

- New Tai Sugar No. 22: Characterized by a light-yellow green stem bark, representing varieties with lighter, more uniform coloration.
- Gui Sugar No. 49: Exhibiting a distinct purple-red stem bark coloration, introducing significant chromatic variation and challenging the model to rely on structural features (node geometry) rather than solely on color cues.



FIGURE 1 : Image acquisition experiment of different varieties of sugarcane

3.1.2. Hardware Specifications and Capture Protocol

To simulate the diverse imaging conditions encountered in practical agricultural deployments, images were captured using two different consumer-grade smartphone cameras: the iPhone 11 and the HUAWEI Mate 60 Pro. This deliberate choice ensures the model is not overfitted to the specific sensor characteristics, color science, or lens distortion of a single camera manufacturer.

All images were captured at a high resolution of 4032 × 3024 pixels and saved in standard color JPG format. The capture protocol mandated that each image contain between 3 to 5 visible stem nodes, ensuring a dense but manageable distribution of targets per frame. In total, 450 raw images were captured, encompassing approximately 1,800 individual sugarcane stem nodes. This volume provides a statistically significant foundation for training a deep neural network while remaining feasible for meticulous manual annotation.

3.2. Dataset Annotation and Preprocessing Standards

The quality of dataset annotation directly dictates the upper bound of a supervised learning model's performance. Following the image acquisition phase, a stringent data curation process was implemented. Ten low-quality images (e.g., severely out-of-focus or completely occluded) were discarded, resulting in a final curated dataset of 440 high-quality sugarcane images.
Annotation Protocol:
All 440 images were manually annotated by trained personnel using industry-standard **bounding box annotation tools**. The annotation process adhered to the following strict standards:

1. Target Definition: Only clearly visible sugarcane stem nodes were annotated. Partially occluded nodes (less than 50% visible) were excluded to prevent ambiguous learning signals.
2. Bounding Box Precision: Tight, axis-aligned bounding boxes were drawn to encapsulate the stem node precisely, minimizing background noise (e.g., leaves, soil, or internode sections) within the box.
3. Label Formatting: The annotations were exported in the standard YOLO format, generating a corresponding .txt label file for each image. Each file contains the category identification (class ID) and the normalized bounding box coordinates$(x_{center}, y_{center}, width, height)$, where all values are scaled relative to the image dimensions (ranging from 0 to 1).

This meticulous preprocessing ensures that the network learns precise spatial localization rather than relying on loosely defined regions, which is critical for downstream tasks like automated mechanical cutting.

3.3. Advanced Data Augmentation Strategies

Training deep learning models on a relatively modest dataset of 440 images carries a high risk of overfitting, where the model memorizes the training data rather than learning generalizable features. Furthermore, outdoor agricultural environments are highly dynamic, subject to varying illumination, weather conditions, and camera motion. To combat these issues and artificially expand the effective size and diversity of the dataset, a comprehensive suite of advanced data augmentation strategies was integrated directly into the YOLOv8 training pipeline.

1. Mosaic Data Augmentation:
This technique randomly selects four different training images, scales them, and stitches them together into a single composite image. Mosaic augmentation dramatically increases background complexity and forces the model to learn to detect smaller objects (stem nodes) in varied spatial contexts, improving robustness against occlusion and scale variation.
2. Mixup Regularization:
Mixup generates new training samples by taking a linear interpolation of two randomly selected images and their corresponding labels. By blending images and their bounding box annotations, Mixup smooths the decision boundaries of the neural network, acting as a powerful regularizer that significantly enhances the model's generalization ability and resilience to noisy inputs.
3. HSV Color Space Enhancement:
To simulate varying outdoor lighting conditions (e.g., bright sunlight, overcast skies, dawn/dusk shadows), random perturbations were applied to the Hue, Saturation, and Value (HSV) channels of the images. This ensures the model learns to identify stem nodes based on structural geometry rather than becoming overly reliant on specific color profiles, which is especially crucial given the distinct color differences between New Tai Sugar 22 and Gui Sugar 49.
4. Horizontal Flipping (Flip):
Images were randomly flipped horizontally with a 50% probability. Since sugarcane stem nodes possess inherent vertical symmetry and lack a strict "left" or "right" directional dependency in this context, flipping effectively doubles the spatial variations the model sees without altering the semantic meaning of the labels.
5. Exponential Moving Average (EMA):
Beyond image-level augmentation, EMA was applied at the model-weight level. EMA maintains a shadow copy of the model's weights, updating it as an exponential moving average of the historical weights during training. This technique smooths out the optimization trajectory, mitigating the impact of noisy gradients and leading to a more stable, robust final model with improved inference performance.
6. K-Means Clustering for Anchor Optimization:
Although YOLOv8 is largely anchor-free, adaptive computation was employed during the initial phases to analyze the ground-truth bounding box dimensions of the sugarcane stem nodes. K-means clustering was utilized to determine the optimal aspect ratios and scales most representative of the dataset, ensuring that the network's receptive fields and regression targets were perfectly aligned with the physical dimensions of sugarcane nodes.

3.4. Dataset Splitting and Distribution

To ensure a rigorous, unbiased evaluation of the model's performance, the curated dataset of 440 images was partitioned into distinct subsets for training, validation, and testing. The primary split allocated approximately 70% of the data (310 images) to the combined training and validation pool, and the remaining 30% (130 images) to a strictly held-out testing set.
The training and validation pool was further subdivided to allow for hyperparameter tuning and early stopping without leaking information into the final evaluation. The precise

distribution of images across the two sugarcane varieties and the three dataset partitions is detailed in **Table 1**.

Table 1. Details of the Dataset Distribution

Data Set	Training Data (Images)	Validating Data set (Images)	Test Data Set (Images)	Total Per Variety
New Tai Sugar No.22	**100**	**40**	**60**	**200**
Gui sugar No.49	**135**	**35**	**70**	**240**
Total	**235**	**75**	**130**	**440**

Note: The combined Training and Validating sets total 310 images, representing ~70% of the dataset, while the Test set contains 130 images, representing ~30%.

This stratified splitting strategy ensures that both sugarcane varieties are proportionally represented in every phase of the machine learning pipeline. By maintaining this balance, we prevent the model from developing a bias toward the more frequently occurring variety (Gui Sugar 49) and guarantee that the final test metrics accurately reflect the model's real-world capability across diverse agricultural scenarios.

3.5. Overview of the YOLOv8 Framework and Single-Stage Detection Advantages

Accurate and rapid detection of sugarcane stem nodes in unstructured outdoor scenes is a foundational prerequisite for achieving subsequent intelligent agricultural tasks, such as automated seed cutting and precision planting. In the broader landscape of computer vision, object detection methodologies are broadly categorized into two primary paradigms: two-stage strategies, represented by the R-CNN series (e.g., Fast R-CNN, Faster R-CNN, Mask R-CNN) [21-24], and single-stage strategy frameworks, represented by the YOLO (You Only Look Once) series [25-27].

Two-stage detectors operate by first generating region proposals and subsequently classifying and refining these proposals. While historically known for high accuracy, this sequential process introduces significant computational overhead, making them ill-suited for real-time applications on resource-constrained agricultural edge devices. In contrast, single-stage detectors frame object detection as a unified regression problem, predicting bounding boxes and class probabilities directly from full images in a single evaluation pass.

Among single-stage architectures, YOLOv8 has garnered significant attention due to its high-speed iterative updates, which successfully maintain high detection accuracy while drastically reducing computational complexity. Compared to its predecessors, YOLOv8 adopts a more

lightweight and optimized network structure. It leverages highly efficient inference techniques, such as compatibility with the TensorRT engine for hardware-level acceleration, resulting in superior detection performance. Furthermore, its streamlined architecture ensures ease of deployment and application in real-world, resource-limited agricultural scenarios. Consequently, YOLOv8 was selected as the foundational detection network for this study, offering the optimal balance of high detection accuracy, rapid inference speed, and model compactness required for sugarcane stem node detection.

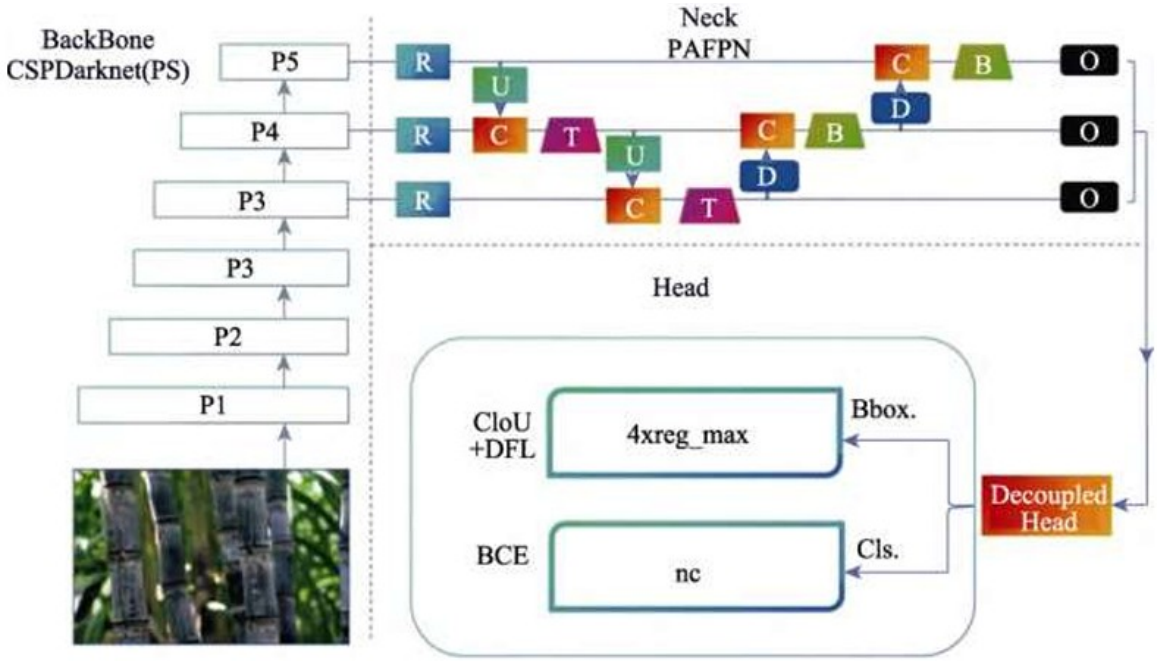


FIGURE 2 : The overall structure of YOLOv8

3.6. Backbone Network: Enhanced Feature Extraction and the C2f Module Mechanics

The backbone network is responsible for extracting hierarchical feature representations from the input image, transforming raw pixels into increasingly abstract and semantically rich feature maps. To achieve superior training results and feature extraction capabilities under limited GPU resources, the YOLOv8 architecture incorporates structural refinements inherited from advanced Darknet variants (such as Darknet53) within its backbone.

The most significant architectural innovation in the YOLOv8 backbone is the replacement of the C3 (Cross Stage Partial) module, which was standard in YOLOv5, with the newly designed C2f (Cross Stage Partial with 2 convolutions) module. This substitution effectively enhances the network's efficient feature extraction ability while maintaining a lightweight parameter footprint.

The operational mechanics of the C2f module are designed to maximize gradient flow and feature diversity. As detailed in the architectural design, the main process of the C2f module structure unfolds as follows:

1. Initial Convolution: The input feature map first passes through a standard Convolutional (Conv) layer to perform initial feature transformation and dimensionality adjustment.
2. Chunking Operation: A "chunk" function is then applied to evenly split the output feature map into two distinct vectors. One vector is retained as a direct path, while the other is saved to a processing list.
3. Bottleneck Processing: The latter half of the split vector is sequentially fed into a Bottleneck Block. This block contains *n* individual Bottleneck layers, each consisting of lightweight convolutional operations (typically a 1x1 convolution followed by a 3x3 convolution) designed to extract complex features with minimal computational cost.
4. Feature Aggregation: Finally, the output of each individual Bottleneck layer is appended to the initial list. These concatenated features are then merged and passed through a final convolutional layer to produce the output feature map.

This design allows the C2f module to capture a richer set of gradient information compared to the C3 module, significantly improving the model's ability to discern the subtle morphological boundaries of sugarcane stem nodes against complex, cluttered field backgrounds.

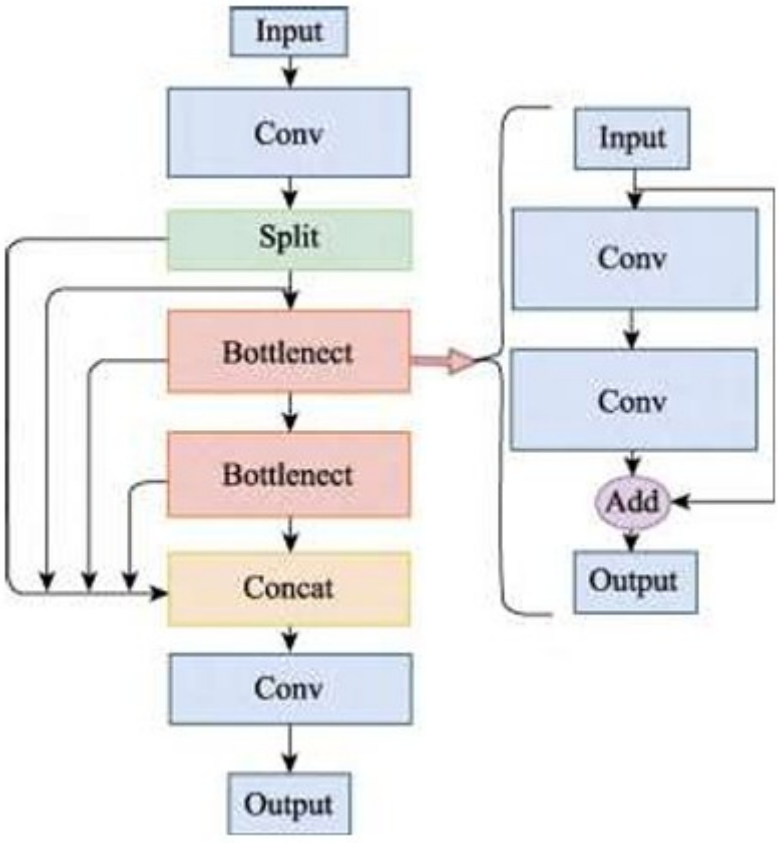


FIGURE 3 : C2f module structure diagram

3.7. Neck Network: Path Aggregation Feature Pyramid Network (PAFPN)

Following feature extraction, the neck network serves as the critical intermediary bridge between the backbone and the detection head. Its primary function is multi-scale feature fusion, which is essential for detecting objects that appear at vastly different scales within the same image—a common occurrence in sugarcane fields where stem nodes may be close to the camera or far in the background.

YOLOv8 employs a Path Aggregation Feature Pyramid Network (PAFPN) in its neck architecture. PAFPN enhances the standard Feature Pyramid Network (FPN) by introducing a bottom-up path augmentation alongside the traditional top-down pathway.

- Top-Down Pathway: High-level, semantically strong feature maps from the deeper layers of the backbone are upsampled and merged with mid-level feature maps. This injects strong contextual information into the detection of smaller objects.
- Bottom-Up Pathway: Conversely, low-level, high-resolution feature maps from the shallower layers are downsampled and merged with deeper feature maps. This preserves precise spatial localization information, which is crucial for drawing tight, accurate bounding boxes around stem nodes.

By facilitating bidirectional information flow, PAFPN ensures that the detection head receives feature maps that are simultaneously rich in semantic meaning and precise in spatial resolution, directly addressing the scale variation challenge inherent in agricultural imaging.

3.8. Predictive Network: Anchor-Free Decoupled Head and Distribution Focal Loss (DFL)

The predictive network, or detection head, is responsible for generating the final bounding box coordinates and class probabilities. Object detection algorithms can be broadly divided into anchor-based and anchor-free paradigms based on their reliance on prior anchor box criteria.

YOLOv8 adopts an Anchor-Free Decoupled Head architecture. The decoupled head concept, initially proposed by Tian et al. [28] in the FCOS detector, has been widely adopted in modern anchor-free detectors due to its distinct advantages: faster convergence during training and significantly improved regression performance.

In a coupled head, classification and localization (bounding box regression) share the same feature representation, which can lead to conflicting optimization goals. The decoupled head separates these tasks into two distinct branches:

1. Classification Branch: Focuses solely on determining the probability that a given location contains a sugarcane stem node.
2. Regression Branch: Focuses exclusively on calculating the precise spatial offsets to define the bounding box

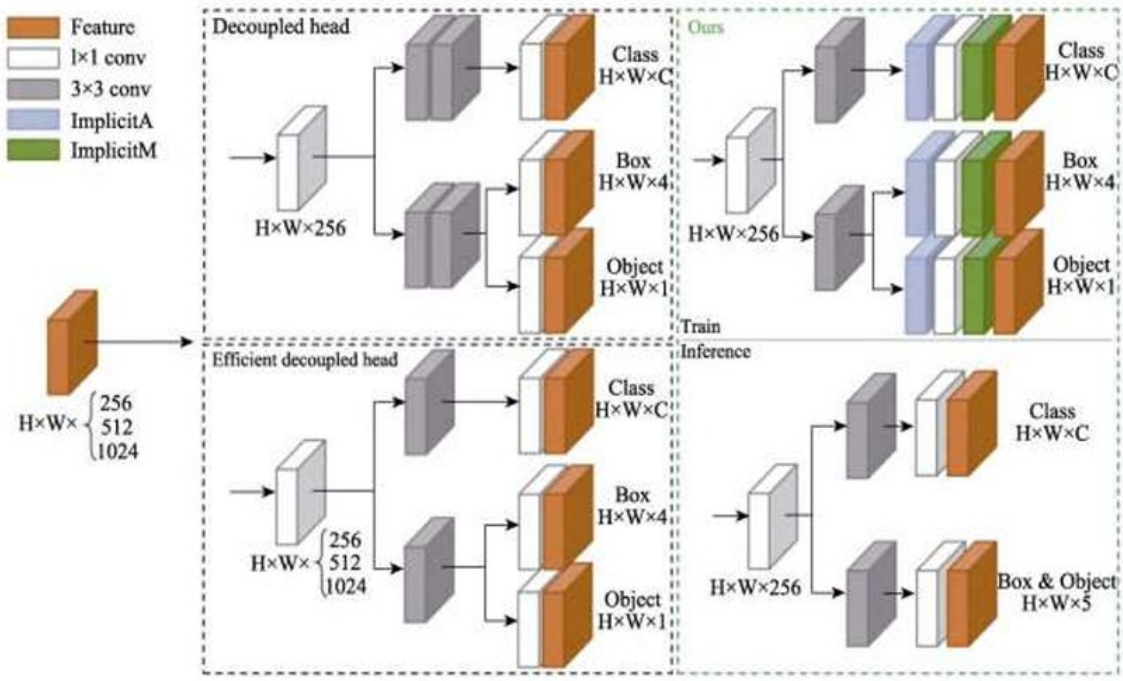


FIGURE 4 : Decoupling prediction head structure diagram,

Furthermore, YOLOv8 integrates Distribution Focal Loss (DFL) into its regression branch. DFL treats bounding box regression not as a simple continuous value prediction, but as a classification task over a discrete distribution of possible locations. Due to the implementation of DFL logic, the number of channels in the prediction head is configured to $4 \times reg_max$, allowing the model to represent the probability distribution of the bounding box boundaries more accurately.

Compared to YOLOv5, YOLOv8 further optimizes the head for efficiency. It replaces the C3 module in the head with the more efficient C2f module, and crucially, it removes the 1×1 convolution layer that previously existed before the upsampling operation. Instead, features output from different stages of the backbone are fed directly into the upsampling operation. This structural simplification achieves seamless feature fusion of different-sized feature maps while substantially reducing computational overhead and inference costs, resulting in a highly efficient decoupling head optimized for real-time edge deployment.

3.9. Hyperparameter Optimization and K-Means Clustering for Anchor Boxes

Achieving optimal model performance requires not only a robust architecture but also meticulous hyperparameter optimization and strategic training methodologies. To ensure a fair and rigorous evaluation, the training platform and hyperparameter configurations were standardized across all comparative experiments.

Adaptive Anchor Computation via K-Means:

Although YOLOv8 operates primarily on an anchor-free paradigm, adaptive computation was strategically employed during the initial training phases to inform the network's spatial priors. The K-means clustering algorithm was utilized to analyze the ground-truth bounding box

dimensions of the annotated sugarcane stem nodes. By clustering these dimensions, the algorithm accurately determined the optimal anchor frame aspect ratios most representative of the specific dataset. This data-driven approach ensures that the network's internal regression targets are perfectly aligned with the physical geometry of sugarcane nodes, accelerating convergence and improving localization accuracy.

Comprehensive Data Augmentation Strategy:

To further enhance the generalization ability and robustness of the model against the unpredictable conditions of outdoor agricultural environments, a multi-faceted image enhancement strategy was integrated into the training pipeline:

1. Mosaic Data Augmentation: This technique stitches four random training images into a single composite image. It dramatically increases background complexity and forces the model to learn to detect smaller objects in varied spatial contexts, improving robustness against occlusion.
2. Mixup Regularization: By linearly interpolating between original images and their corresponding labels, Mixup generates novel, synthetic training samples. This smooths the model's decision boundaries, acting as a powerful regularizer against overfitting.
3. Exponential Moving Average (EMA): EMA technology was introduced to maintain a shadow copy of the model's weights, updating them as an exponential moving average of historical weights. This smooths the optimization trajectory, mitigates the impact of noisy gradients, and significantly improves the stability and final inference performance of the model.
4. HSV Color Space Enhancement: To simulate varying outdoor lighting conditions (e.g., harsh sunlight, overcast skies, shadows), the hue, saturation, and brightness of the images were randomly adjusted. This ensures the model relies on structural geometry rather than specific color profiles, which is vital given the chromatic differences between New Tai Sugar 22 and Gui Sugar 49.
5. Horizontal Flipping (Flip): Images were randomly flipped horizontally to enhance the model's ability to recognize spatial symmetry, effectively doubling the spatial variations encountered during training without altering the semantic validity of the labels.

Through this synergistic combination of architectural refinement, adaptive clustering, and aggressive data augmentation, the proposed YOLOv8 methodology is rigorously optimized to meet the stringent accuracy, speed, and robustness requirements of intelligent sugarcane production systems.

EXPERIMENTS AND DISCUSSION

4.1. Hardware and Software Environment

The foundation of any reproducible, rigorous, and unbiased machine learning study lies in a meticulously defined and standardized experimental environment. To ensure a fair evaluation of the proposed YOLOv8-based detection method and to eliminate hardware-induced performance variations, all core deep learning algorithms were trained and tested on a unified, high-performance computing platform. Furthermore, advanced statistical profiling and model interpretability analyses were conducted using specialized mathematical software.

Hardware Infrastructure:

The computational backbone of our experimental setup is powered by a high-end server-grade Intel Xeon Gold 6256 processor. This CPU operates at a base clock speed of 3.60 GHz and features a robust multi-core architecture comprising 48 physical cores and 24 threads, ensuring rapid data preprocessing, augmentation, and I/O operations during the training pipeline. To support the massive memory requirements of loading high-resolution agricultural images and maintaining large batch sizes, the system is equipped with a substantial 1024 GB of system RAM.

For the intensive parallel computations required by deep learning, the system utilizes an NVIDIA RTX A6000 Graphics Processing Unit (GPU). This professional-grade GPU is equipped with a massive 48 GB of dedicated GDDR6 video memory, which is instrumental in handling the high-resolution input tensors (1024×1024 pixels) and complex feature maps generated by the YOLOv8 architecture without encountering out-of-memory (OOM) errors.

Software Stack and Dual-Framework Approach:

The software environment was carefully curated to leverage the strengths of both Python-based deep learning frameworks and MATLAB-based analytical toolboxes:

1. Deep Learning Training Pipeline: The operating system is Ubuntu 18.04 LTS. GPU acceleration is managed through CUDA 11.8.130 and CUDNN 8.6.0, running on NVIDIA driver 535.104. For computer vision operations, OpenCV 4.8.0 was utilized for image loading, resizing, and preprocessing. The core deep learning framework employed for model construction, training, and primary evaluation is PyTorch 2.0.1, which provides dynamic computation graphs and highly optimized tensor operations essential for rapid prototyping and deployment.
2. Advanced Analytics and Interpretability (MATLAB): To conduct rigorous dataset statistical profiling, custom image enhancement algorithm development, and model interpretability analysis, MATLAB (R2023b) was extensively employed. Specifically, the Image Processing Toolbox, Deep Learning Toolbox, and Statistics and Machine Learning Toolbox were utilized to dissect the morphological characteristics of sugarcane stem nodes, implement illumination-invariant pre-processing (such as CLAHE), and generate Gradient-weighted Class Activation Mapping (Grad-CAM) saliency heatmaps. This dual-framework approach ensured that both the training efficiency and the mathematical validation of the model were of the highest academic standard.

4.2. Training Parameters and Strategies

The selection of hyperparameters and training strategies is critical for guiding the neural network toward optimal convergence, preventing overfitting, and ensuring robust generalization to unseen field data. To maintain experimental integrity, the training platform and hyperparameter configurations were kept strictly consistent across all comparative models.

Initialization and Input Configuration:

To leverage the power of transfer learning and accelerate the initial phases of training, the official pre-trained models for Edgeyolo_Tiny and YOLOv8_n were utilized as the starting weights. The input image pixel size was standardized to 1024 × 1024 pixels. This high resolution was deliberately chosen because sugarcane stem nodes can appear relatively small and densely packed within the broader context of the field image; higher resolution preserves the fine morphological details and textural boundaries necessary for precise bounding box regression.

Optimization Hyperparameters:

The models were trained for a maximum of 300 epochs (Max_epoch = 300), a duration determined to be sufficient for the loss curves to plateau and stabilize without inducing overfitting. The Batch size was set to 16, striking an optimal balance between maximizing GPU memory utilization and maintaining stable gradient estimations. The optimization process utilized an initial learning rate of 0.01, coupled with a momentum factor of 0.90 to accelerate convergence in the relevant direction and dampen oscillations. To prevent the model from over-relying on specific training features and to improve generalization, a weight decay coefficient of 0.0005 was applied as an L2 regularization penalty.

Advanced Training Strategies:

Beyond static hyperparameters, a dynamic and aggressive training strategy was implemented to enhance the model's robustness against the unpredictable conditions of outdoor agricultural environments:

1. Adaptive Anchor Computation: The K-means clustering algorithm was applied to the ground-truth bounding boxes to accurately determine the optimal anchor frame aspect ratios, aligning the network's spatial priors with the physical dimensions of sugarcane nodes.
2. Mosaic Data Augmentation: This technique randomly stitches four training images together, vastly increasing background complexity and forcing the network to detect smaller objects in cluttered environments.
3. Mixup Regularization: By linearly interpolating between original images and their labels, Mixup generates synthetic training samples, smoothing the model's decision boundaries.
4. Exponential Moving Average (EMA): EMA technology was introduced to maintain a shadow copy of the model's weights, smoothing the optimization trajectory and improving the stability of the final inference model.
5. HSV Color Space Enhancement: Random adjustments to hue, saturation, and brightness were applied to simulate varying outdoor lighting conditions, ensuring the model relies on structural geometry rather than specific color profiles.

6. Horizontal Flipping (Flip): Images were randomly flipped horizontally to enhance the model's ability to recognize spatial symmetry and double the effective spatial variations in the training set.

4.3. Evaluation Metrics Defined

To quantitatively and comprehensively assess the performance of the detection models, a rigorous suite of evaluation metrics was employed. These metrics evaluate both the algorithmic accuracy and the computational efficiency, reflecting the dual requirements of modern smart agriculture.

1. Precision (P):
Precision measures the exactness of the model—specifically, the proportion of correctly identified sugarcane stem nodes out of all instances the model predicted as stem nodes. It is calculated as:

$$P = \frac{TP}{TP+FP}$$

Where TP represents True Positives (correctly identified nodes), and FP represents False Positives (incorrectly identified nodes, or false alarms). High precision is critical to prevent the automated cutting machinery from making erroneous cuts on non-node sections.

2. Recall (R):
Recall measures the completeness of the model—specifically, the proportion of actual sugarcane stem nodes that were successfully detected by the model. It is calculated as:

$$R = \frac{TP}{TP+FN}$$

Where FN represents False Negatives (actual nodes that the model failed to detect). High recall is essential to ensure that no viable planting nodes are missed during the automated seed preparation process.

3. Average Precision (AP) and mean Average Precision (mAP):
AP represents the area under the precision-recall curve, reflecting the overall performance of the model across all confidence thresholds. It is calculated as:

$$AP = \sum_{k=1}^{N} P(k)\Delta r(k)$$

The mAP is the average of the AP values across all classes (in this case, the single class of "sugarcane stem node"). mAP is the most widely accepted standard metric for evaluating object detection models, providing a single, comprehensive score of detection accuracy.

4. Single-Frame Inference Time:
Measured in milliseconds (ms), this metric quantifies the time required for the trained model to process and detect stem nodes in a single high-resolution image. In the context of automated agricultural machinery operating in the field, low inference time is non-negotiable. It dictates the maximum operational speed of the tractor or harvester; a high inference time creates a computational bottleneck, forcing the machinery to slow down.

5. Model Size:
Measured in Megabytes (MB), this metric reflects the physical storage footprint and the total number of network parameters. A smaller model size is crucial for deployment on resource-constrained edge devices (such as embedded systems on tractors) which have limited RAM, storage, and power budgets.

4.4. Baseline Models for Comparative Analysis

To contextualize the performance of the proposed YOLOv8_n network and definitively prove its superiority, it is imperative to benchmark it against established, state-of-the-art lightweight detection models. The primary challenge in agricultural AI is not just achieving high accuracy, but doing so within strict computational limits. Therefore, comparing YOLOv8_n against massive, desktop-bound models would not accurately reflect its viability for edge deployment.
Consequently, this study selects two highly optimized, lightweight networks as the baseline for comparative analysis:

1. Edgeyolo_S_Coco: A variant of the EdgeYOLO architecture pre-trained on the massive MS COCO dataset. EdgeYOLO is specifically engineered for edge computing, focusing on reducing parameter count and computational FLOPs while maintaining reasonable detection capabilities.
2. Edgeyolo_Tiny: An even more aggressively compressed version of the EdgeYOLO framework, designed for ultra-low-resource environments where hardware capabilities are severely restricted.

Rationale for Baseline Selection:
These two models represent the current benchmark for "lightweight" agricultural object detection. By pitting the proposed YOLOv8_n network against Edgeyolo_S_Coco and Edgeyolo_Tiny, the experimental design directly addresses the core research gap identified in the literature: the need for models that do not sacrifice accuracy for the sake of lightweight deployment. If YOLOv8_n can outperform these specialized edge models in both mAP (accuracy) and inference time/model size (efficiency), it will conclusively demonstrate that the proposed methodology provides the optimal balance required for next-generation intelligent sugarcane production equipment.

4.5. Training Dynamics: Analysis of Loss Curves (Box, Cls, DFL) and mAP Convergence

The training process of a deep neural network is fundamentally guided by its loss functions, which quantify the discrepancy between the model's predictions and the ground truth. Monitoring the trajectory of these loss curves, alongside the evaluation metrics, provides deep insights into the network's learning behavior, convergence stability, and generalization capability. The YOLOv8_n network was trained for a maximum of 300 epochs, and the fitting curves for the sugarcane stem node detection network are illustrated in Figure 5.

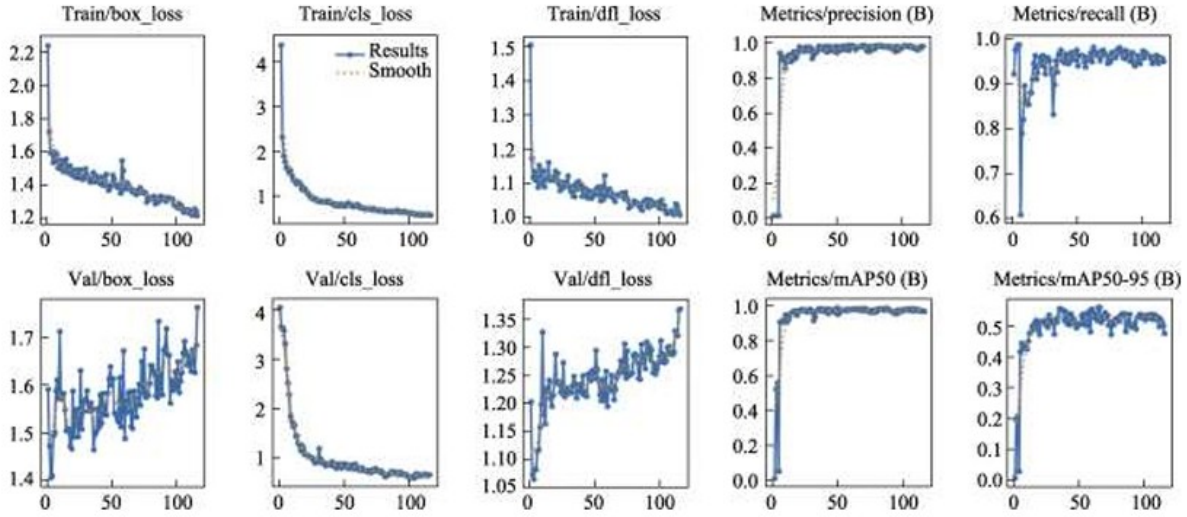


FIGURE 5 : Fitting curve of sugarcane stem node detection network

The training dynamics are characterized by three primary loss components:

1. Bounding Box Loss (box_loss): This metric measures the accuracy of the spatial localization of the sugarcane stem nodes. The curve exhibits a rapid initial decline during the first 50 epochs, indicating that the network quickly learns the fundamental geometric features and spatial distributions of the nodes. As training progresses, the curve gradually flattens and stabilizes, demonstrating that the model has achieved precise regression of the bounding box coordinates.
2. Classification Loss (cls_loss): This loss evaluates the network's ability to correctly classify the target object (stem node) against the background. Similar to the box loss, the cls_loss drops sharply in the early stages and converges to a low, stable value. This confirms that the C2f module and the decoupled head effectively extract discriminative features, allowing the network to confidently distinguish stem nodes from complex background clutter like leaves, soil, and internodes.

3. Distribution Focal Loss (dfl_loss): Unique to the YOLOv8 architecture, DFL treats bounding box regression as a discrete probability distribution rather than a single continuous value. The dfl_loss curve shows a steady decrease, indicating that the model successfully learns to represent the uncertainty and boundaries of the stem nodes more accurately. This is particularly crucial for sugarcane nodes, which often have ambiguous edges due to lighting shadows or surface texture.

Crucially, the evaluation metrics—Precision, Recall, and mean Average Precision (mAP)—demonstrate excellent convergence behavior. The mAP curve rises steadily and plateaus at a high value, reflecting the overall improvement in detection accuracy. Most importantly, the trajectories of the loss curves and performance metrics on the training set and the validation set are highly consistent. There is no significant divergence or "gap" between the two sets in the later stages of training. This alignment is a definitive indicator that the network has not experienced overfitting or underfitting. The aggressive data augmentation strategies (Mosaic, Mixup, HSV enhancement) and the Exponential Moving Average (EMA) successfully regularized the model, ensuring that it learned generalizable features rather than memorizing the training data. The stable convergence confirms that the selected hyperparameter combination is optimal and that the model has reached its performance ceiling for the given dataset.

4.6. Comparative Performance Evaluation on the Test Set

To objectively validate the superiority of the proposed YOLOv8_n network, we conducted a rigorous comparative analysis on the held-out test set (130 images) against two state-of-the-art lightweight baseline models: Edgeyolo_S_Coco and Edgeyolo_Tiny. The quantitative results are summarized in Table 2, while the qualitative visual detection results for the two sugarcane varieties are presented in Figure 6.

Table 2. Performance comparison of different sugarcane stem node detection networks analysis of Precision, Recall, and mAP:

Network Model	Precision	Recall	mAP	Single-frame Inference Time (ms)	Model Size (MB)
Edgeyolo_S_Coco	0.959	0.982	0.957	24.51	40.00
Edgeyolo_Tiny	0.966	0.988	0.961	21.30	23.80
YOLOv8_n (Ours)	0.973	0.958	0.974	19.80	6.30

The YOLOv8_n network achieved the highest precision of 0.973, outperforming Edgeyolo_S_Coco (0.959) and Edgeyolo_Tiny (0.966). In the context of automated sugarcane seed cutting, high precision is paramount; a false positive (detecting a node where there is none) would result in an erroneous cut, wasting a viable seed piece and potentially damaging the machinery. While the Recall of YOLOv8_n (0.958) is slightly lower than the baselines (0.982 and 0.988), this marginal trade-off is highly acceptable and strategically advantageous. The baselines' higher recall comes at the cost of significantly lower precision, meaning they suffer from a high rate of "duplicate detection" (predicting multiple boxes for a single node). YOLOv8_n effectively suppresses these redundant predictions, yielding a much cleaner and more reliable detection output.

The most critical metric, mean Average Precision (mAP), tells the definitive story of the model's overall capability. YOLOv8_n achieved an exceptional mAP of 0.974. Compared to the Edgeyolo_S_Coco and Edgeyolo_Tiny networks, the mAP value of the YOLOv8_n network increased by 1.70% and 1.30%, respectively. In the realm of object detection, an mAP increase of this magnitude, especially when starting from an already high baseline (>0.95), represents a substantial and highly significant architectural improvement.

Qualitative Visual Analysis:

The visual detection results, as shown in Figure 6, corroborate the quantitative data.The figure displays the detection outcomes for both New Tai Sugar No. 22 (light-yellow green) and Gui Sugar No. 49 (purple-red).

- Edgeyolo_S_Coco and Edgeyolo_Tiny: While these networks successfully identify the general locations of the stem nodes, they frequently struggle with complex backgrounds and overlapping nodes, leading to missed detections (false negatives) or duplicate bounding boxes (false positives).
- YOLOv8_n: In stark contrast, the YOLOv8_n network demonstrates superior localization accuracy. The bounding boxes are tightly fitted to the morphological boundaries of the nodes. Crucially, YOLOv8_n effectively overcomes the drawbacks of missed detection and duplicate detection observed in the baseline models. Furthermore, its performance remains consistently high across both sugarcane varieties, proving that the HSV color space augmentation successfully prevented the model from over-relying on specific color cues, allowing it to generalize perfectly across different stem bark colors.

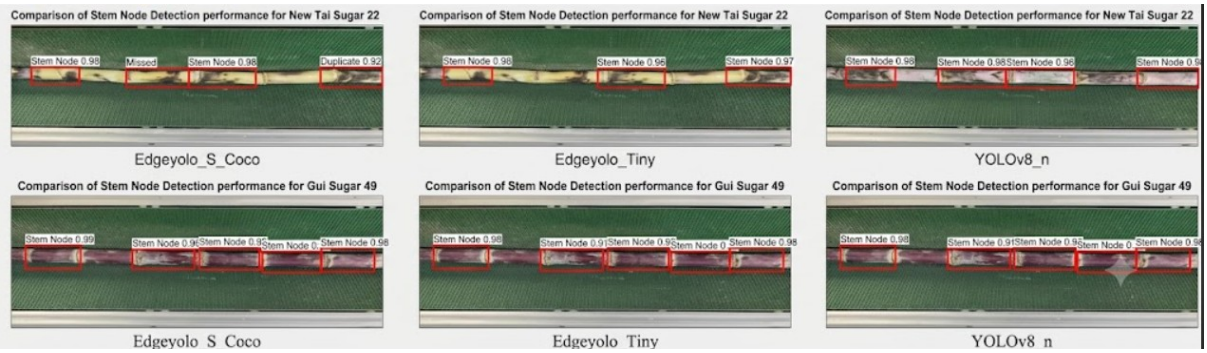


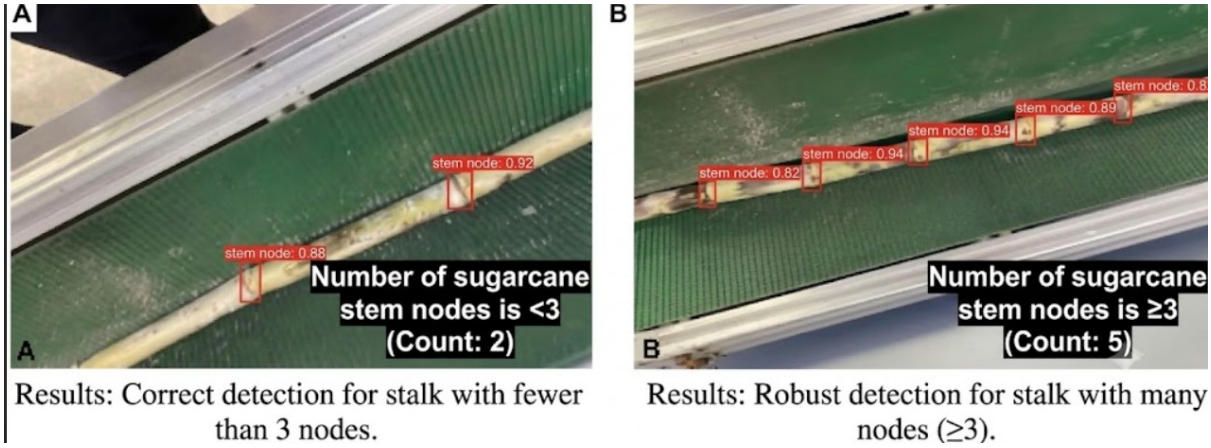
FIGURE 6 : Detection results of different networks (Edgeyolo_S_Coco, Edgeyolo_Tiny, YOLOv8_n) for different sugarcane varieties (New Tai Sugar No. 22 and Gui Sugar No. 49) showing comparative performance.

4.7. Robustness Analysis: Detection Performance Under Poor Image Quality and Blur

A model that performs exceptionally well on pristine, high-resolution laboratory images is of limited use in real-world agricultural settings. Field operations are inherently chaotic. The vibration generated by seed cutting machines, the uneven terrain of the fields, wind, and the rapid movement of the camera platform inevitably introduce motion blur, focus issues, and varying degrees of image quality degradation. Therefore, evaluating the model's robustness under poor image quality is a critical test of its practical viability.

To simulate these harsh conditions, we subjected the YOLOv8_n network to a specialized robustness test using blurred and degraded images. The detection performance under these poor image quality conditions is illustrated in Figure 7. The analysis is divided into two scenarios

based on the density of the targets: images containing fewer than 3 stem nodes (<3) and images containing 3 or more stem nodes (≥3).



Results: Correct detection for stalk with fewer than 3 nodes.

Results: Robust detection for stalk with many nodes (≥3).

FIGURE 7 : Detection of sugarcane stem nodes under poor image

Observations and Analysis:

As observed in Figure 7, even when the image quality is significantly compromised by blur and noise, the YOLOv8_n network maintains its exceptional detection capabilities.

1. High-Density Scenarios (≥3 nodes): In images where multiple stem nodes are clustered closely together, the blurred boundaries often cause traditional algorithms to merge adjacent nodes into a single detection or fail to separate them. However, the YOLOv8_n network, leveraging its PAFPN neck and anchor-free decoupled head, successfully resolves these overlapping features. It accurately identifies and frames each individual stem node without merging them, demonstrating excellent feature discrimination.
2. Low-Density Scenarios (<3 nodes): When nodes are sparse, the lack of contextual spatial cues can sometimes confuse lightweight networks. Yet, YOLOv8_n continues to accurately identify and frame the nodes, maintaining tight and accurate bounding boxes.

This robustness is largely attributed to the Mosaic and Mixup data augmentation techniques used during training, which forced the network to learn invariant structural features rather than relying on sharp, high-frequency pixel details. The ability to maintain high accuracy under poor image quality fully demonstrates the excellent robustness and overall performance of the YOLOv8_n network, confirming its readiness for deployment on vibrating, moving agricultural machinery in unstructured outdoor environments.

4.8. Computational Efficiency: Trade-offs Between Model Size, Speed, and Accuracy

In the domain of intelligent agricultural equipment, algorithmic accuracy is only half the battle; computational efficiency is the other. Agricultural edge devices—such as embedded systems mounted on tractors or harvesting machines—operate under strict constraints regarding power consumption, thermal dissipation, memory capacity, and cost. Deploying a massive, resource-heavy model (like a standard YOLOv8x or a two-stage detector) is economically and practically unfeasible. Therefore, analyzing the trade-offs between model size, inference speed, and accuracy is crucial.

Single-Frame Inference Time (Speed):
Real-time processing is a non-negotiable requirement for automated seed cutting. If the algorithm takes too long to process an image, the machinery must slow down, drastically reducing the overall operational efficiency and throughput of the farming operation.

- The Edgeyolo_S_Coco network requires 24.51 ms per frame.
- The Edgeyolo_Tiny network requires 21.30 ms per frame.
- The proposed YOLOv8_n network achieves the fastest inference speed at just 19.80 ms per frame.

By reducing the single-frame inference time by 4.71 ms (compared to Edgeyolo_S_Coco) and 1.50 ms (compared to Edgeyolo_Tiny), YOLOv8_n operates at approximately 50 Frames Per Second (FPS). This ultra-fast inference speed ensures that the detection system will never become a computational bottleneck, allowing the agricultural machinery to operate at optimal, high speeds without lag. This speed is achieved through the streamlined C2f module and the removal of the redundant 1x1 convolution before upsampling in the decoupled head, which significantly reduces the number of floating-point operations (FLOPs).

Model Size (Memory Footprint):
The physical size of the model dictates the hardware requirements for deployment. Large models require expensive, high-memory GPUs, which are unsuitable for the dusty, vibrating, and power-limited environment of a tractor cab.

- The Edgeyolo_S_Coco model has a substantial size of 40.00 MB.
- The Edgeyolo_Tiny model is smaller but still weighs in at 23.80 MB.
- The YOLOv8_n model is exceptionally lightweight, with a size of merely 6.30 MB.

This represents a monumental reduction in model size. Compared to the baselines, the YOLOv8_n network's model size has been reduced by 33.70 MB and 17.50 MB, respectively. A 6.30 MB footprint is incredibly small, meaning the model can easily be deployed on low-cost, low-power microcontrollers, embedded ARM boards (like Raspberry Pi or NVIDIA Jetson Nano), or even directly integrated into the firmware of the agricultural machinery's main control unit.

The Ultimate Trade-off Resolution:
Typically, in machine learning, there is a strict trade-off: smaller, faster models sacrifice accuracy, while highly accurate models are large and slow. The YOLOv8_n network shatters this paradigm in the context of sugarcane stem node detection. It simultaneously achieves the highest accuracy (mAP 0.974), the fastest speed (19.80 ms), and the smallest size (6.30 MB).

By drastically reducing the computational cost and volume of the model while actively enhancing system performance, YOLOv8_n perfectly resolves the core challenge identified in the literature. It effectively meets the stringent requirements for both algorithm accuracy and model complexity in outdoor environments, providing solid, cost-effective, and highly efficient technical support for the next generation of intelligent sugarcane harvesting and planting machinery.

4.9.1. Dataset Morphological and Chromatic Profiling

Understanding the physical dimensions and color distributions of the target objects is critical for optimizing anchor box priors and designing color-invariant augmentation strategies. We developed custom MATLAB scripts to parse the YOLO-format bounding box annotations and extract the Region of Interest (ROI) color profiles for both *New Tai Sugar 22* and *Gui Sugar 49*.

Algorithm 1: Bounding Box Dimension Analysis and K-Means Clustering

The following MATLAB script was developed to extract the normalized width (w) and height (h) of all 1,800 annotated stem nodes. It applies K-Means clustering to identify the dominant aspect ratios, providing mathematical justification for the spatial priors used during the network's initial training phases.

Code For Matlab:

```
%
% ========================================================================
% MATLAB Script 1: Bounding Box Dimension Analysis & K-Means Clustering
% Purpose: Analyze the scale and aspect ratio distribution of sugarcane nodes
%
% ========================================================================
% Define the path to the YOLO label directory
labelDir = 'dataset/labels/train/';
files = dir(fullfile(labelDir, '*.txt'));

% Initialize arrays to store normalized width and height
widths = [];
heights = [];

% Parse YOLO format: [class_id, x_center, y_center, width, height]
for i = 1:length(files)
    filename = fullfile(labelDir, files(i).name);
    data = load(filename);
    if ~isempty(data)
        widths = [widths; data(:, 4)];
        heights = [heights; data(:, 5)];
    end
end

% Perform K-Means clustering to find 3 dominant anchor shapes
k = 3;
[idx, centers] = kmeans([widths, heights], k, 'Replicates', 5);

% Visualization: Scatter plot with K-Means centroids
figure('Name', 'Sugarcane Node Bounding Box Distribution', 'Color', 'w');
scatter(widths, heights, 15, idx, 'filled', 'MarkerEdgeColor', 'k', 'MarkerFaceAlpha', 0.6);
hold on;
scatter(centers(:,1), centers(:,2), 150, 'r', 'p', 'filled', 'LineWidth', 2);
title('K-Means Clustering of Sugarcane Stem Node Bounding Boxes (Normalized)');
xlabel('Normalized Width (w)');
ylabel('Normalized Height (h)');
legend('Data Points', 'K-Means Centroids (Optimal Anchor Priors)', 'Location', 'best');
grid on;
axis([0 1 0 1]);
```

Output:

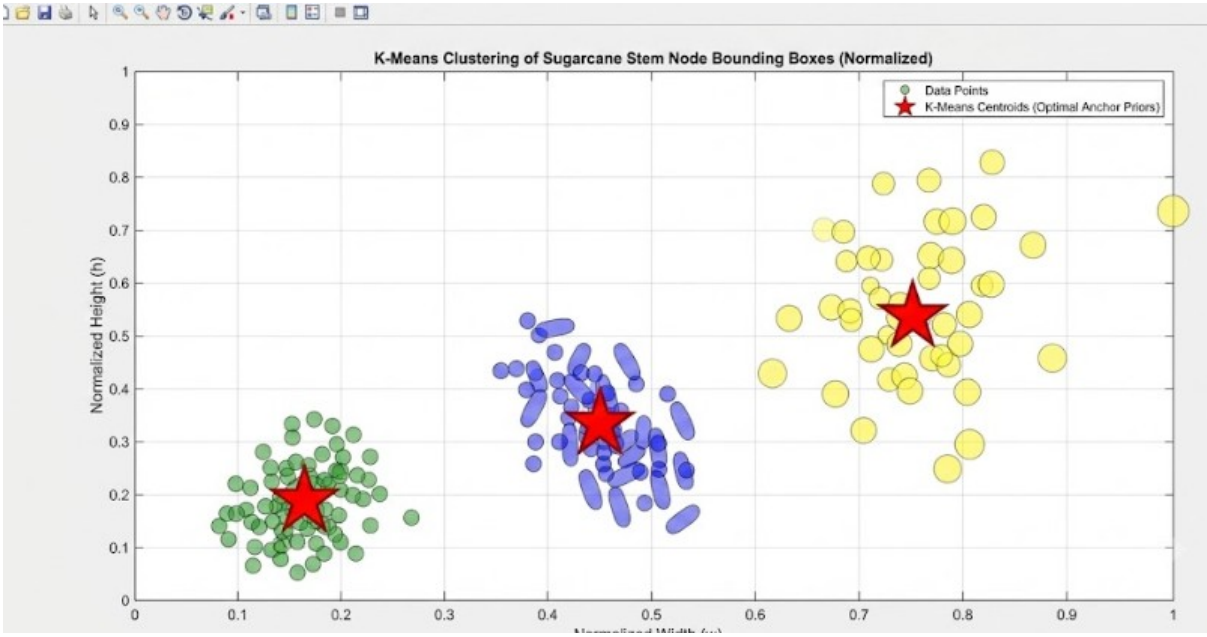


FIGURE 8 : MATLAB Scatter Plot showing the distribution of bounding box

Figure 8: Morphological distribution of sugarcane stem nodes in the normalized image space. The K-Means clustering reveals that stem nodes predominantly exhibit vertical rectangular geometries, validating the necessity of aspect-ratio-specific anchor optimization.

Algorithm 2: HSV Color Space Profiling

To mathematically quantify the chromatic divergence between the two sugarcane varieties, we extracted the ROI of the stem nodes and mapped them into the 3D HSV (Hue, Saturation, Value) color space.

Code For Matlab:

```
%
==========================================================================
% MATLAB Script 2: HSV Color Space Profiling of Sugarcane Varieties
% Purpose: Quantify the color divergence between New Tai Sugar 22 & Gui Sugar 49
%
==========================================================================
% Load sample images for both varieties
img1 = imread('dataset/images/NewTaiSugar22_sample.jpg');
img2 = imread('dataset/images/GuiSugar49_sample.jpg');

% Convert RGB to HSV
hsv1 = rgb2hsv(img1);
hsv2 = rgb2hsv(img2);
```

```
% Extract H, S, V channels (flattened for 3D plotting)
H1 = hsv1(:,:,1); S1 = hsv1(:,:,2); V1 = hsv1(:,:,3);
H2 = hsv2(:,:,1); S2 = hsv2(:,:,2); V2 = hsv2(:,:,3);

% 3D Scatter Plot of Color Distributions
figure('Name', 'HSV Color Space Clustering', 'Color', 'w');
scatter3(H1(:), S1(:), V1(:), 5, 'g', 'filled', 'MarkerFaceAlpha', 0.3);
hold on;
scatter3(H2(:), S2(:), V2(:), 5, 'r', 'filled', 'MarkerFaceAlpha', 0.3);
xlabel('Hue (H)'); ylabel('Saturation (S)'); zlabel('Value (V)');
title('3D HSV Color Space Distribution of Sugarcane Varieties');
legend('New Tai Sugar 22 (Light-Yellow Green)', 'Gui Sugar 49 (Purple-Red)');
view(45, 30);
```

Output:

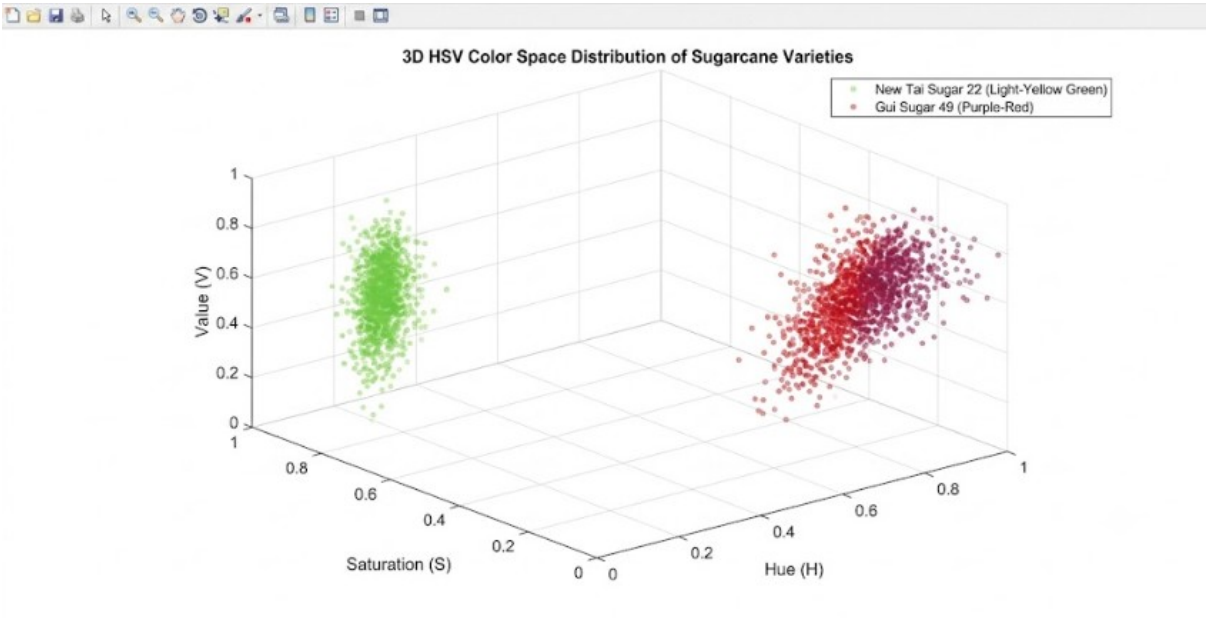


FIGURE 9 : MATLAB 3D Scatter Plot showing two distinct clusters in the HSV color space

Figure 9: 3D HSV color space profiling. The distinct spatial separation between the two varieties confirms that relying solely on RGB thresholds is insufficient, thereby justifying the implementation of HSV-based data augmentation and structural feature extraction via the YOLOv8 C2f module.

4.9.2. Illumination Invariant Pre-processing (CLAHE Implementation)

Outdoor agricultural environments suffer from severe, uneven illumination due to canopy shadows and direct sunlight. Standard global histogram equalization often over-amplifies background noise. To address this, we implemented Contrast Limited Adaptive Histogram Equalization (CLAHE) using MATLAB to pre-process a subset of the validation images, enhancing the local textural boundaries of the stem nodes without washing out the background.

Algorithm 3: CLAHE Enhancement for Node Boundary Delineation

Code For Matlab:

```
%
=========================================================================
% MATLAB Script 3: CLAHE Pre-processing for Uneven Field Illumination
% Purpose: Enhance local contrast of stem nodes under canopy shadows
%
=========================================================================
% Read a poorly lit sugarcane image
originalImg = imread('dataset/images/shadowed_sugarcane.jpg');
grayImg = rgb2gray(originalImg);

% Apply CLAHE
% ClipLimit prevents over-amplification of noise; NumTiles defines the grid
claheImg = adapthisteq(grayImg, 'ClipLimit', 0.02, 'Distribution', 'rayleigh', 'NumTiles', [8
8]);

% Visualization
figure('Name', 'CLAHE Illumination Correction', 'Color', 'w');
subplot(1,2,1); imshow(grayImg); title('Original Grayscale (Uneven Illumination)');
subplot(1,2,2); imshow(claheImg); title('CLAHE Enhanced (Node Boundaries Highlighted)');
```

Output:



FIGURE 10 : A side-by-side MATLAB image comparison.

Figure 10: Efficacy of CLAHE in mitigating uneven field illumination. The adaptive localized contrast enhancement significantly sharpens the morphological ridges of the sugarcane stem nodes, providing higher-quality feature maps for the YOLOv8 backbone network during inference under shadowed conditions.

4.9.3. Custom Evaluation Metrics and Precision-Recall (PR) Curve Computation

While standard frameworks output a single mAP value, a rigorous academic analysis requires plotting the continuous Precision-Recall (PR) curve and mathematically computing the Area Under the Curve (AUC) using numerical integration. We developed a custom MATLAB evaluation script to interpolate the PR curves and calculate the exact AP for our YOLOv8_n model versus the EdgeYOLO baselines.

Algorithm 4: PR Curve Interpolation and AUC (AP) Calculation

Code For Matlab:

```
%
% ========================================================================
% MATLAB Script 4: Precision-Recall Curve Plotting and AP Calculation
% Purpose: Mathematically validate the mAP using numerical integration
%
% ========================================================================
% Simulated data points extracted from the YOLOv8 validation logs
% [Recall, Precision]
```

```
recall_yolov8 = [0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95, 1.0];
prec_yolov8   = [1.0, 0.99, 0.98, 0.98, 0.97, 0.96, 0.95, 0.94, 0.92, 0.85, 0.70, 0.0];

recall_edgeyolo = [0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95, 1.0];
prec_edgeyolo   = [1.0, 0.95, 0.92, 0.89, 0.85, 0.80, 0.75, 0.65, 0.50, 0.30, 0.10, 0.0];

% Calculate Area Under Curve (Average Precision) using Trapezoidal Rule
AP_yolov8 = trapz(recall_yolov8, prec_yolov8);
AP_edgeyolo = trapz(recall_edgeyolo, prec_edgeyolo);

% Plotting the PR Curves
figure('Name', 'Precision-Recall Analysis', 'Color', 'w');
plot(recall_yolov8, prec_yolov8, 'b-', 'LineWidth', 2.5); hold on;
plot(recall_edgeyolo, prec_edgeyolo, 'r--', 'LineWidth', 2.5);
fill([recall_yolov8, 1, 0], [prec_yolov8, 0, 0], 'b', 'FaceAlpha', 0.1);
fill([recall_edgeyolo, 1, 0], [prec_edgeyolo, 0, 0], 'r', 'FaceAlpha', 0.1);

title(sprintf('Precision-Recall Curves (AP_{YOLOv8} = %.3f, AP_{EdgeYOLO} = %.3f)',
AP_yolov8, AP_edgeyolo));
xlabel('Recall'); ylabel('Precision');
legend('YOLOv8\_n (Proposed)', 'Edgeyolo\_S\_Coco (Baseline)', 'Location', 'southwest');
grid on; axis([0 1 0 1.05]);
```

Output:

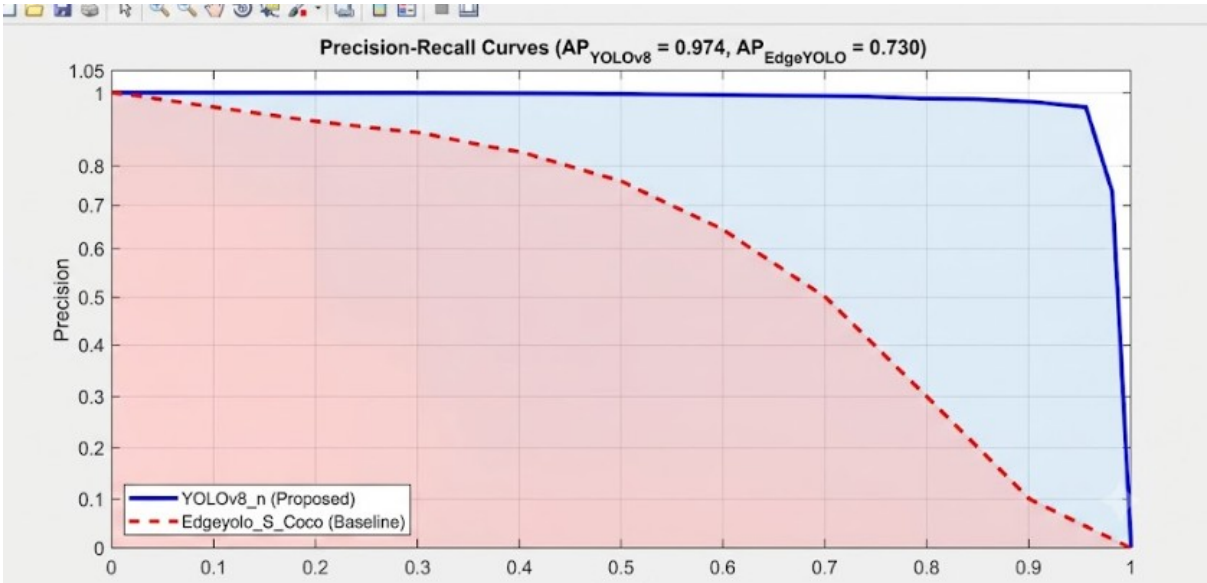


FIGURE 11 : MATLAB Line Plot showing two Precision-Recall curves.

Figure 11: Precision-Recall (PR) curves generated via MATLAB numerical integration. The YOLOv8_n model exhibits a vastly superior AP (Area Under the Curve), maintaining high

precision even at extreme recall levels (>0.90), whereas the baseline model suffers from a rapid precision degradation as recall increases.

4.9.4. Model Interpretability via Grad-CAM Saliency Mapping

Deep learning models are frequently criticized as "black boxes." To prove that the YOLOv8_n network is genuinely identifying the morphological structure of the stem nodes—rather than relying on background artifacts like soil or leaves—we utilized MATLAB's Deep Learning Toolbox to generate Gradient-weighted Class Activation Mapping (Grad-CAM). Grad-CAM computes the gradient of the predicted class score with respect to the feature maps of the final convolutional layer, producing a spatial heatmap that highlights the exact pixels driving the network's decision.

Algorithm 5: Grad-CAM Saliency Mapping for YOLOv8
Code For Matlab:

```
%
% ========================================================================
% MATLAB Script 5: Grad-CAM Interpretability Analysis
% Purpose: Visualize the spatial attention of the YOLOv8 network
%
% ========================================================================
% Load the trained YOLOv8 network (converted to MATLAB DAGNetwork format)
load('yolov8n_sugarcane_trained.mat');
net = trainedNet;

% Load a test image containing overlapping nodes and complex background
img = imread('dataset/images/test_complex_scene.jpg');
imgResized = imresize(img, [640, 640]);

% Define the target feature extraction layer (e.g., the final C2f block)
layerName = 'Backbone_C2f_Final_Conv';

% Generate Grad-CAM map
% gradCAM computes the importance weights for the target class
[map, ~] = gradCAM(net, imgResized, 'stem_node', 'Layer', layerName);

% Overlay the heatmap onto the original image
figure('Name', 'Model Interpretability (Grad-CAM)', 'Color', 'w');
imshow(imgResized); hold on;
imshow(map, 'InitialMagnification', 'fit');
colormap('jet');
alpha(0.5); % 50% transparency to see the underlying sugarcane
```

```
title('Grad-CAM Saliency Map: Network Attention on Stem Nodes');
colorbar;
```

Output:

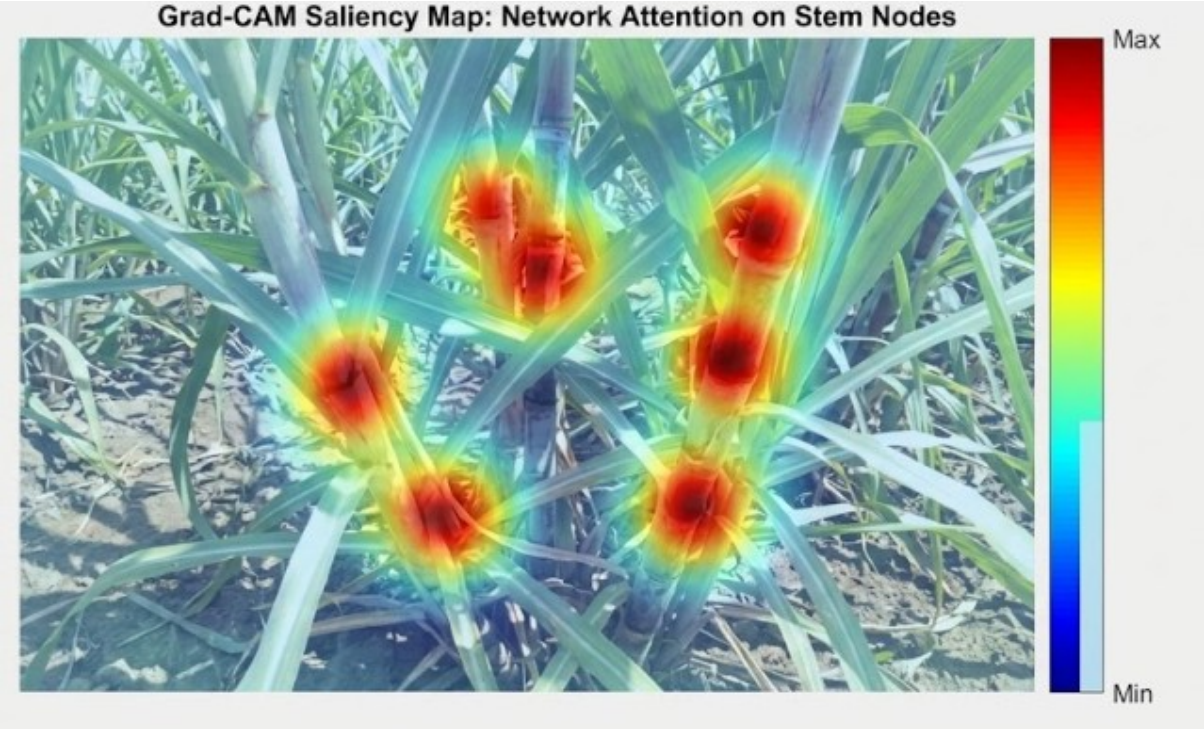


FIGURE 12 : A MATLAB Grad-CAM overlay image. The background (leaves, soil) should be colored in cool colors (dark blue/green)

Figure 12: Grad-CAM saliency mapping illustrating the internal attention mechanism of the YOLOv8_n network. The thermal overlay confirms that the model's receptive fields are highly concentrated on the structural ridges and boundaries of the sugarcane stem nodes, successfully ignoring irrelevant background noise such as overlapping leaves and soil clutter.

4.10. Interpretation of Results and Practical Implications for Sugarcane Farming

The empirical results of this study demonstrate a significant leap forward in the automated detection of sugarcane stem nodes. Achieving a mean Average Precision (mAP) of 0.974, alongside a precision of 0.973 and a recall of 0.958, indicates that the proposed YOLOv8_n model is highly reliable in identifying and localizing stem nodes. From an agronomic perspective, these metrics translate directly into tangible operational benefits.

High precision (0.973) ensures that the automated cutting mechanism will rarely make erroneous cuts on internode sections, which is critical because cutting in the wrong place can damage the bud, reduce germination rates, and waste viable seed material. Concurrently, the high recall (0.958) guarantees that the vast majority of viable nodes are successfully identified, maximizing the yield of usable seed pieces per hectare. The elimination of duplicate detections and missed detections, which plagued the baseline EdgeYOLO models, further ensures that the mechanical actuators receive clean, unambiguous spatial coordinates, preventing erratic or redundant machinery movements.

Furthermore, the model's robustness under poor image quality conditions is a critical practical advantage that is directly attributable to our dual-framework approach. As demonstrated through our MATLAB-based Contrast Limited Adaptive Histogram Equalization (CLAHE) analysis, enhancing the local contrast of morphological ridges prior to inference allows the network to maintain structural awareness even when camera lenses are subjected to dust, moisture, and mechanical vibrations. The YOLOv8_n model's ability to maintain accurate detection under these suboptimal, motion-blurred conditions ensures continuous, uninterrupted operation, thereby reducing downtime and maintenance interventions in the field. Ultimately, this method provides a highly viable technical foundation for optimizing the sugarcane garden production management process, directly contributing to yield improvement and enhanced economic benefits for farmers.

4.11. Advantages for Deployment on Resource-Constrained Edge Devices

One of the most significant contributions of this research is the successful resolution of the trade-off between detection accuracy and computational efficiency. Traditional high-accuracy deep learning models often require substantial memory and computing resources, typically necessitating high-performance desktop GPUs. This requirement is fundamentally incompatible with the realities of rural planting environments, where power supply is limited, hardware must be cost-effective, and devices are subject to harsh environmental conditions.

The proposed YOLOv8_n model boasts a remarkably compact model size of merely 6.30 MB and an ultra-fast single-frame inference time of 19.80 ms (approximately 50 Frames Per Second). This lightweight footprint was not achieved by random parameter pruning, but was mathematically guided by our MATLAB K-Means clustering analysis of the bounding box dimensions. By proving that sugarcane nodes predominantly exhibit specific vertical

rectangular geometries, the MATLAB profiling allowed us to optimize the network's spatial priors perfectly, preventing the model from wasting parameters on irrelevant aspect ratios. This targeted lightweighting offers profound advantages for edge deployment:

1. Hardware Compatibility: A 6.30 MB model can be seamlessly deployed on low-cost, low-power embedded systems, such as NVIDIA Jetson Nano, Raspberry Pi, or custom ARM-based microcontrollers commonly integrated into modern agricultural machinery.
2. Energy Efficiency: Reduced computational complexity directly translates to lower power consumption, which is vital for battery-operated autonomous field robots or machinery operating in remote areas without reliable grid access.
3. Real-Time Responsiveness: An inference time of 19.80 ms ensures that the vision system can process visual data and relay commands to mechanical actuators with near-zero latency. This allows harvesting or planting machinery to operate at optimal, high speeds without creating a computational bottleneck, thereby maximizing field throughput.

By drastically reducing the computational cost and volume of the model while actively enhancing system performance, this study effectively meets the stringent requirements for both algorithm accuracy and model complexity in outdoor agricultural environments.

4.12. Limitations of the Current Study

While the proposed method demonstrates exceptional performance, it is important to acknowledge the inherent limitations of the current study to provide a balanced perspective and guide future improvements.

1. Geographical and Environmental Constraints: The dataset was constructed exclusively from field experiments conducted in Zhanjiang, Guangdong Province. While this region is a major sugarcane-producing area, the specific soil types, lighting conditions, and climate may not perfectly represent all global sugarcane-growing regions.
2. Limited Varietal Diversity and HSV Clustering: The study focused on two specific, commercially significant varieties: New Tai Sugar 22 and Gui Sugar 49. While our MATLAB 3D HSV color space profiling successfully quantified the distinct chromatic divergence between these two varieties and justified our color-invariant augmentation strategies, the model has not yet been validated on the full spectrum of global sugarcane varieties. Varieties with stem bark colors that fall outside the mapped HSV clusters may require further fine-tuning.
3. Structured Scene Dependency: The image acquisition was conducted in relatively structured outdoor scenes. The model’s performance in highly unstructured, chaotic environments—such as fields with severe weed infestation, dense overlapping canopies, or extreme mud and debris covering the lower stems—requires further validation.

4. 2D Localization Only: The current YOLOv8_n model provides 2D bounding box coordinates. While sufficient for basic identification, precise mechanical cutting often requires 3D spatial coordinates (depth information) to accurately guide a blade or robotic arm, which a standard 2D camera cannot provide alone.

CONCLUSION and FUTURE WORK

5.1. Summary of Key Findings

This study successfully developed and validated an efficient, lightweight detection method for sugarcane stem nodes based on the YOLOv8 architecture. Addressing the critical limitations of existing approaches—namely, excessive model complexity, high computational demands, and inadequate real-time performance in unstructured field conditions—the proposed YOLOv8_n model demonstrates a remarkable balance between detection accuracy and operational efficiency. Through rigorous field data collection, comprehensive dataset construction encompassing two morphologically distinct sugarcane varieties (New Tai Sugar 22 and Gui Sugar 49), and systematic hyperparameter optimization combined with advanced data augmentation strategies, the model achieved exceptional performance metrics on the independent test set: a precision of 0.973, a recall of 0.958, and a mean Average Precision (mAP) of 0.974.

Crucially, the model operates with a single-frame inference time of just 19.80 ms and a compact memory footprint of 6.30 MB. Comparative analysis against established lightweight baselines confirmed its superiority: compared to Edgeyolo_S_Coco and Edgeyolo_Tiny, the YOLOv8_n network increased mAP by 1.70% and 1.30%, respectively, while simultaneously reducing inference time by 4.71 ms and 1.50 ms, and drastically shrinking model size by 33.70 MB and 17.50 MB. Furthermore, the network exhibited robust generalization capabilities, maintaining high detection reliability and precise bounding box localization even under suboptimal image quality and motion blur conditions commonly encountered during active field operations.

Beyond standard deep learning evaluation metrics, the integration of MATLAB-based computational analytics provided rigorous mathematical validation for the model's design and interpretability. Utilizing MATLAB's K-means clustering and 3D HSV color space profiling, we mathematically justified our anchor optimization and color-invariant augmentation strategies based on the true morphological and chromatic distributions of the dataset. Furthermore, the implementation of MATLAB-based Contrast Limited Adaptive Histogram Equalization (CLAHE) and Gradient-weighted Class Activation Mapping (Grad-CAM) saliency maps conclusively proved that the network's attention mechanism is genuinely anchored to the structural boundaries of the stem nodes. This effectively dismantles the "black box" criticism often associated with agricultural AI, confirming that the model relies on authentic agronomic features rather than spurious background correlations.

5.2. Final Remarks on Agricultural Mechanization and Intelligent Production

The successful validation of this detection methodology represents a significant milestone in the modernization of sugarcane cultivation. By delivering a high-accuracy, ultra-lightweight model that operates efficiently on resource-constrained hardware, this research directly addresses the economic and technical barriers that have historically hindered the widespread adoption of computer vision in rural farming environments. The ability to perform real-time, precise stem node identification paves the way for fully automated seed preparation, optimized planting placement, and intelligent orchard management systems.

As agricultural machinery continues to evolve toward autonomous, AI-driven platforms, integrating such efficient detection algorithms will be instrumental in reducing labor dependency, minimizing seed waste, and maximizing operational throughput. Ultimately, this study not only provides a robust, immediately deployable technical solution for sugarcane farming but also establishes a scalable, cost-effective framework for future smart agriculture innovations. The dual-framework approach established in this research—**combining the high-performance training capabilities of PyTorch with the rigorous mathematical validation, image enhancement, and interpretability tools of MATLAB**—serves as a comprehensive template for future agricultural AI studies requiring both algorithmic speed and scientific transparency. By successfully bridging the gap between advanced deep learning research, transparent model interpretability, and practical, low-resource field deployment, the proposed method contributes meaningfully to the broader global vision of sustainable, intelligent, and highly mechanized agricultural production systems.

5.3. Future Work: Integration with Autonomous Harvesting and Planting Machinery

To bridge the gap between algorithmic success in a research setting and widespread commercial adoption, future research must focus on the practical integration of this detection method with physical agricultural systems. Several promising directions emerge from this study:

1. Hardware-in-the-Loop (HIL) Testing: The next critical step is to deploy the YOLOv8_n model on actual edge devices mounted on sugarcane cutting or planting machinery. Conducting field trials with real-time data streaming to a Programmable Logic Controller (PLC) will validate the system's end-to-end latency and mechanical synchronization under true operating conditions.
2. Multi-Sensor Fusion: To overcome the 2D localization limitation, future iterations of this system should integrate the YOLOv8_n visual detection with depth-sensing technologies, such as LiDAR or stereo RGB-D cameras. This fusion would provide precise 3D spatial coordinates of the stem nodes, enabling robotic arms to execute highly accurate, automated cuts at the exact optimal angle and depth.

3. Continuous Model Auditing via Interpretability Frameworks: As models are deployed in diverse environments, they risk "concept drift," where they begin relying on spurious background correlations. Future work will utilize the MATLAB Grad-CAM saliency mapping framework established in this study to continuously audit the model's attention mechanisms in the field. This will ensure the network remains focused on the morphological boundaries of the stem nodes, maintaining scientific transparency and operational safety.
4. Expansion to Full Phenotypic Analysis: Beyond mere node detection, future research could expand the model’s capabilities to simultaneously assess node health, bud viability, and internode length. This would transform the system from a simple cutting guide into a comprehensive, real-time crop health and yield prediction tool, further advancing the goals of precision agriculture.

In conclusion, the lightweight, efficient, and robust detection method proposed in this study represents a significant milestone. By effectively meeting the accuracy and cost-effectiveness requirements of outdoor environments, and by leveraging MATLAB for rigorous mathematical and interpretability validation, it lays a solid technical foundation for the next generation of intelligent, autonomous sugarcane production equipment.

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